

Optimal Portfolio Strategy to Control Maximum Drawdown

The Case of Risk-based Active Management with Dynamic Asset Allocation

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Abstract

Draw-down losses from a previously reached maximum portfolio wealth level, is an important risk measure for investment management. In this study, we present a discrete trading strategy to directly control a portfolio's maximum percentage of drawdown within a target level while maximizing the portfolio's long term growth rate.

Modifying the continuously rebalancing models proposed by Grossman and Zhou (1993) ^[1] and Cvitanic and Karatzas (1995) ^[2], we define the loss control target as a *Rolling Economic Drawdown (REDD)* with a constant look-back window progressing in time. Additionally, investor's risk aversion in the power law wealth utility function is specified as complement of the maximum loss limit to construct a portfolio's risk-return efficient frontier.

We test the dynamic strategy using data of three broad asset class indexes: S&P 500 Total Return Index (*SPTR*), Barclays Capital 20+ Year US Treasury Bond Index (*TLT*) and Dow-Jones UBS Commodity Total Return Index (*DJUBS*), with 3-month US Treasury Bill as the risk-free asset. Over a test period of the past 20 years (1992-2011), a simplified risk-based out-of-sample dynamic asset allocation portfolio is robust against variations in capital market expectation inputs, and out-performs the in-the-sample calibrated model portfolio and traditional asset allocation portfolios significantly.

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Executive Summary

Draw-down, the percentage of loss from a previous peak wealth level, is a critical risk measure for investment management. Originated from the high water-mark practice of the hedge fund industry, the portfolio's maximum draw-down metric has profound implication for manager incentives, performance measurement, and the broad objective of client retention for any investment management practice.

Without an explicit loss control mechanism, traditional passive asset allocation portfolios experienced large drawdowns during the 2008-2009 financial crises. In this study, we propose an active quantitative trading strategy to directly control portfolio's drawdown within a target level while maximizing long term portfolio wealth growth rate.

With 40-year S&P 500 Index data including the post-crisis 2009-2011 market recovery, we can demonstrate that the classical mathematical framework of continuous drawdown control by Grossman and Zhou (1993) did not achieve optimality under discrete trading extension. This is due to the model's implied one-step trend-following nature, but sizing position with a long term anchored drawdown look-back.

We introduce a new concept of *Rolling Economic Drawdown (REDD)* to define a maximum loss limit. Counting investor's opportunity loss at risk free rate (e.g. the 3-month T-Bill yield), *REDD* looks back the portfolio wealth history during a rolling time window of a fixed length. By specifying investor's risk aversion as the complement of drawdown loss limit, we are able to derive explicit formula of optimal dynamic asset

allocation weights for one or multiple risky assets. They depend on capital market expectations of long term expected Sharpe ratios, volatilities and correlation coefficients.

Further, an adaptive position scaling by ratio $\frac{\delta - REDD}{1 - REDD}$ is the key to control portfolio

maximum drawdown below a target level δ . This is our active management proposal of a new *Rolling Economic Drawdown-Controlled Optimal Portfolio Strategy (REDD-COPS)*.

REDD-COPS are calibrated with historical data of three broad asset class indexes: *S&P 500 Total Return Index (SPTR)*, *Barclays Capital 20+ Year US Treasury Bond Index (TLT)* and *Dow-Jones UBS Commodity Total Return Index (DJUBS)* as risky assets up to six decades but with focus on the past 20+ years. We use a rolling time window of *one year* to define *REDD*, considering investor's liquidity and behavioral constraints, and an expected time scale of market decline cycle before recovery.

Systematic active management through *REDD-COPS* is applied progressively in back-test: first only to position sizing with calibrated constant parameter inputs, secondly as an *out-of-sample* risk-based dynamic asset allocation scheme. Compared to traditional Stock/Bond asset allocation benchmarks, such as a 60/40 Balanced Portfolio, Minimum Variance Portfolio (*MVP*) or a Leveraged Risk Parity Portfolio (*RPP*) over recent 20+ years, the calibrated *REDD-COPS* achieves *dominance* on risk-return efficient frontiers. Adding commodities index (*DJUBS*) as a third asset in the active scheme further improves diversification benefit. Out-performing all calibrated models, the *out-of-sample* risk-based *REDD-COPS* achieved around 5% excess annualized return over traditional risk-based asset allocation portfolios with similar maximum drawdowns. For a 20-year test period (1992-2011), the following table lists the performance comparison.

	Single Index Buy & Hold			SPTR-TLT Fixed Allocation Portfolio			SPTR-TLT-DJUBS Risk-based REDD-COPS		
	SPTR	TLT	DJUBS	Balanced (60-40)	MVP	60% Levered RPP	15% REDD Target	20% REDD Target	25% REDD Target
Annualized Return	7.81%	8.93%	5.65%	8.79%	9.04%	11.29%	11.06%	13.71%	16.45%
Annualized Std Deviation	15.00%	11.37%	15.23%	9.46%	8.40%	13.35%	10.29%	13.98%	17.90%
Sharpe Ratio	0.301	0.495	0.154	0.580	0.683	0.598	0.754	0.745	0.734
Average REDD	7.21%	5.64%	8.01%	4.31%	3.68%	5.45%	3.69%	5.01%	6.41%
Max REDD	46.76%	21.51%	54.48%	26.68%	20.51%	26.04%	14.65%	19.68%	24.85%
Max Drawdown	50.95%	21.40%	54.26%	28.63%	19.09%	28.16%	13.98%	19.00%	24.10%
Maximum Exposure	100%	100%	100%	100%	100%	160%	292.3%	396.8%	507.9%
Average Exposure	100%	100%	100%	100%	100%	160%	143.3%	195.2%	250.3%
Ending Multiple	4.501	5.533	3.001	5.389	5.641	8.489	8.143	13.065	21.014

Unlike the common active management using technical or fundamental analysis, active trading scheme in *risk-based REDD-COPS* makes simple theoretical assumptions about capital markets: long term un-correlated asset classes exist; their returns statistically follow random walk and they have long term stable market price of risk. Despite actual market's deviations from the simplified premises in the last 20 years (1992-2011), the *risk-based REDD-COPS* demonstrate *robustness* of out-performance in back test. It is an active asset allocation framework for practical portfolio management.

Besides the benefit of a clear mandate to construct suitable client portfolios, *REDD-COPS* methodology can be applied to design invest-able securities, such as principal-guaranteed investment product, target risk asset allocation *ETF*, and target-date mutual fund with glide path, etc. With active mechanisms derived solely from risk statistics and risk target, the current proposal of *REDD-COPS* demonstrates the prowess of risk based investment management: driven by active risk control, rather than performance chasing, an investment portfolio will out-perform consistently, for the long term.

Acronyms and Symbols

<i>MPT</i>	Modern Portfolio Theory
<i>MVO</i>	Mean Variance Optimization
<i>MVP</i>	Minimum Variance Portfolio
<i>RPP</i>	Risk Parity Portfolio
<i>EM</i>	Economic Maximum (portfolio wealth in history)
<i>REM</i>	Rolling Economic Max
<i>EDD</i>	Economic Drawdown
<i>REDD</i>	Rolling Economic Drawdown
<i>DD-COPS</i>	Draw-down Controlled Optimal Portfolio Strategy
<i>REDD-COPS</i>	Rolling Economic Draw-down Controlled Optimal Portfolio Strategy
<i>H</i>	Rolling time window length in <i>REDD</i> definition
<i>W</i>	Portfolio Wealth Level
γ	Complement of Risk Aversion in the power law investor utility function
δ	Target level of portfolio maximum draw-down, maximum loss limit
s, t, τ	Discrete time variables
<i>R</i>	Long term annualized return
σ	Volatility (evaluated from annualized standard deviation)
ρ	Correlation coefficient
r_f	Average risk free interest rate (e.g. 3-month <i>T-Bill</i> yield) for compounding
x	Portfolio weight
λ	Market price of risk, defined through Sharpe ratio
h	Look back time window for volatility calculation

1. Introduction

Drawdown is a critically important risk measure for investment management. The common definition of drawdown of a portfolio is the percentage loss of current wealth (W_t) from a prior all-time high (M_t): $DD_t = 1 - W_t / M_t$. The portfolio's downside risk of a prolonged drawdown matters not only to the investors' financial well-being, but also to the investment manager's business survival in an immediate term.

Collecting performance based fee, usually at 20% of annual profit, hedge fund managers have been known for the high watermark practice. If the portfolio ends a year that is lower in value than any of the previous year, the manager will not get any performance based compensation. Also drawdown causes account terminations or redemptions. Losing account assets undermines any investment business or even lead to its demise. When it comes to tolerating large drawdowns for a separate managed account, there is no or little regard to whether the strategy is valid in long run with high expected risk adjusted return – it simply cannot survive to that point!

Maximum drawdown challenges a client's financial and psychological tolerance. According to Chekhlov *et al* (2005)^[3], 50% drawdown is unlikely to be tolerated in any average account, and an account may be closed if drawdown breaches 20% or has lasted over two years. For passive index investors, financial markets have been tough: maximum drawdown over 50% occurred for both the *Dow Jones Industrial Average (DJIA)* and the *S&P 500 Index* during the recent 2008-2009 financial crisis. A 50% drawdown, as pointed out by Zhou and Zhu (2009)^[4], however, is 90% probable to happen over a century even if the stock markets are simply modeled as random walk.

Despite being widely used, diversification through passive asset allocation was not effective to avoid large drawdowns. During a market crisis, risky asset classes can exhibit the “contagion” effect: highly correlated losses across the board lead to large drawdowns. Markowitz (1952)^[5] modern portfolio theory (*MPT*) and Mean Variance Optimization (*MVO*) methodology defined risk as return standard deviation, a path-independent statistical attribute. Without an explicit mechanism to control maximum drawdown, it

was not uncommon for a traditional balanced (60% stock + 40% bond) portfolio suffering maximum drawdown loss of 30% during the 2008-2009 financial crises.

Grossman and Zhou (1993) ^[1] pioneered the mathematical frame-work of portfolio optimization under the dynamic floor constraint to control maximum drawdown, extending the constant floor portfolio optimization by Black and Perold (1987) ^[6] – the basis of the theory and practice of Constant Portion Portfolio Insurance (CPPI). In an award-winning Ph.D. thesis work, Grossman and Zhou (1993) approached the problem with *Expected Utility Theory* and defined portfolio optimality as maximizing long term growth rate in power law wealth utility function ($U = W^\gamma / \gamma$). The model assumed continuously rebalancing between a risk free asset and single risky asset, which has random walk return dynamics. Their drawdown calculation accounted the economic decay of portfolio value at the risk free rate r_f ⁴. An *Economic Drawdown (EDD)* was defined by $EDD(t) = 1 - W_t / EM(t)$, where an *Economic Max (EM)* since inception is calculated as $EM(t) = \max_{0 \leq s \leq t} \{(1 + r_f)^{t-s} W_s\}$. The continuous *Drawdown-Controlled Optimal Portfolio Strategy (DD-COPS)* has the portfolio fraction allocated to single risky asset as:

$$x_t = \left(\frac{\lambda / \sigma + \frac{1}{2}}{1 - \delta \cdot \gamma} \right) \cdot \left[\frac{\delta - EDD(t)}{1 - EDD(t)} \right] \quad (1)$$

where δ is the drawdown limit, $(1 - \gamma)$ is the investor risk aversion, and $\lambda = (R - r_f) / \sigma$ is the long term expected Sharpe ratio of the risky asset (R and σ are its long term expected return and volatility). The rest of the portfolio is allocated to the risk free asset. The risky asset allocation has a leverage factor $\left(\frac{\lambda / \sigma + \frac{1}{2}}{1 - \gamma \cdot \delta} \right)$, and is scaled by a timing ratio $\frac{\delta - EDD(t)}{1 - EDD(t)}$, which adaptively controls drawdown. The case of $\delta = 100\%$ gives Merton's (1971) ^[7]

⁴ We take r_f as the average realized risk free rate over a period of length $(t - s)$ and write the compounding factor as $(1 + r_f)^{t-s}$ in *Economic Max* definition. When the risk free rate is changing, we will use

$\prod_{i=s}^t (1 + r_i \cdot \Delta t)$ to compound, where r_i denotes the risk free interest rate during i 'th discrete time interval Δt .

unconstrained optimal portfolio leverage $x = \mu / [(1 - \gamma)\sigma^2]$ where drift $\mu = R - r_f + 1/2 \cdot \sigma^2$ in a continuous random walk model. Optimal leverage from Kelly's criterion formulae $x = \mu / \sigma^2$ is a further special case that an investor has a myopic logarithmic utility function with $\gamma = 0$. Cvitanic and Karatzas (1995)^[2] extended the continuous optimal drawdown control strategy to a case of multiple risky assets. Under discrete time interval rebalance and leverage-limited linear constraints, He (1999)^[8] conducted market data test and obtained piecewise parabola type risk-return efficient frontier.

However, Klass and Nowicki (2005)^[9] countered that a discrete implementation can result in the loss of portfolio optimality. The discrete trading of equation (1) uses updated current portfolio value, but the allocation takes full effect with a finite Δt time delay. Loss of portfolio optimality is due to the finite delay and an anchored long term drawdown look-back. Market cycle from decline to recovery can be much longer than the discrete rebalance frequency. Compared to a continuous rebalancing, discrete trading underweights risky assets during a market downward spiral. Due to long term memory effect of the drawdown control, less exposure to risky asset can cascade into the rebound phase of the market cycle, leading to lower long term accumulated returns. The recent 2007 to 2011 market cycle provides us the opportunity to test the loss of optimality problem. With S&P 500 Index as the risky asset, Appendix 1 gives a data-based demonstration.

2. Analytical Formulation

Economic Drawdown (EDD) in Grossman and Zhou (1993)'s model, reflects an idealistic mental accounting from sophisticated investors: how much better off if they have exited the risky asset completely at a retrospective *perfect* time in history, when a risk free rate compounded historical high was achieved. However, not all investors invested or had memory since time zero – there are portfolio *inception difference* among investors. There are also *liquidity constraints*: not all investors can exit at a perfect time. Hedge fund's initial 1-year lock-up and quarterly redemption window, mutual fund minimum holding period or redemption penalty are examples of restrictions. Practically, at time of a market cycle bottom, using a drawdown reference lower than *Economic Max (EM)* can improve performance as a *forward looking* market timing mechanism.

We propose an alternative to the anchored time window (since inception) for drawdown calculation: a constant *rolling time window*. Define a *Rolling Economic Max (REM)* at time t , looking back at portfolio wealth history for a rolling window of length H :

$$REM(t, H) = \max_{t-H \leq s \leq t} [(1+r_f)^{t-s} W_s] \quad (2)$$

Rolling Economic Max (REM) with a reset feature can be alternatively defined as:

$$REM(t, H) = \begin{cases} \max_{0 \leq s \leq H} [W_{t-s} (1+r_f)^s], & t - \tau \leq H \\ \max[W_t, W_{t-H} (1+r_f)^H], & t - \tau > H \end{cases} \quad \text{where} \quad \tau(t) = \max_{s < t} (s | REM_s = W_s) \quad (3)$$

Equation (3) is the same as (2) to define *REM* for most of the time, but it compounds portfolio value since H period ago at risk-free rate if it has not been renewed by the portfolio value over time H . Equation (2) is used later unless pointed out otherwise.

From *REM*, the *Rolling Economic Draw-down (REDD)* is consequently defined as:

$$REDD(t, H) = 1 - \frac{W_t}{REM(t, H)} \quad (4)$$

Intuitively, a drawdown look-back period H somewhat shorter than or similar to the market decline cycle is the key to achieve optimality. Substituting *EDD* with a lower *REDD* in equation (1), we have higher risky asset allocation to improve portfolio return during a market rebound phase. In the examples followed, we'll use $H=1$ year throughout.

Our second change to the original continuous models is about investor risk profile characterization in a dynamic asset allocation framework. Equation (1) has two risk tolerance parameters: drawdown control target δ and risk aversion complement γ . Gross and Zhou (1993) did not make a direct connection between them. A straightforward conjecture is that a more risk-averse investor should have a lower drawdown loss tolerance. We propose in this study simply $\gamma = \delta$ to use the dynamic floor percentage as the risk aversion parameter in power utility function. In comparison, Kelly's formulae ($x = \mu/\sigma^2$) implies 100% drawdown tolerance ($\delta = 1$) whereas assuming logarithmic wealth utility ($\gamma = 0$), the conservative limit of power law utility functions.

We propose a *Rolling Economic Drawdown – Controlled Optimal Portfolio Strategy (REDD-COPS)*: a dynamic allocation policy between risky asset(s) and a risk free asset, such that the long term wealth growth rate is maximized under a dynamic constraint $REDD \leq \delta$. The dynamic portfolio fraction allocated to single risky asset is:

$$x_t = \text{Max} \left\{ 0, \left(\frac{\lambda/\sigma + \frac{1}{2}}{1 - \delta^2} \right) \cdot \left[\frac{\delta - REDD(t, H)}{1 - REDD(t, H)} \right] \right\} \quad (5)$$

The risk free asset (e.g. 3-month *T-Bill*) accounts for the rest of portfolio allocation $x_f = 1 - x_t$. In discrete trading, equation (5) disallows shorting risky asset but has the portfolio all in risk free asset for any out-of-control time when $REDD > \delta$. Larger expected Sharpe ratio λ , smaller volatility σ of the risky asset, or a higher drawdown tolerance δ , leads to higher or leveraged level of investments in the risky asset. Maximum weighting in single risky asset is $(\lambda/\sigma + \frac{1}{2}) \cdot \delta / (1 - \delta^2)$, realized when $REDD = 0$.

For two risky assets in *REDD-COPS*, dynamic asset allocation weights are:

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{1 - \rho^2} \begin{bmatrix} (\lambda_1 + \frac{1}{2}\sigma_1 - \rho \cdot (\lambda_2 + \frac{1}{2}\sigma_2)) / \sigma_1 \\ (\lambda_2 + \frac{1}{2}\sigma_2 - \rho \cdot (\lambda_1 + \frac{1}{2}\sigma_1)) / \sigma_2 \end{bmatrix} \cdot \text{Max} \left[0, \frac{1}{1 - \delta^2} \cdot \left(\frac{\delta - REDD}{1 - REDD} \right) \right] \quad (6)$$

The portion of risk free asset is $x_f = 1 - x_1 - x_2$. It is obvious that return correlation efficient ρ between two risky assets impacts position size and leverage in the dynamic asset allocation process. When risky assets have positive Sharpe ratios ($\lambda_{1,2} > 0$), no shorting in any risky asset needs $\rho \leq 0$ or $\rho < (\lambda_1 + \frac{1}{2}\sigma_1) / (\lambda_2 + \frac{1}{2}\sigma_2) < 1/\rho$ ($\rho > 0$). In Appendix 2, we derive equation (6), and the explicit asset allocation formula for three risky assets case.

3. Historical Calibration

Traditional asset allocation process needs to estimate asset class returns, standard deviations and correlations as input parameters to the portfolio optimization process. One can calculate these fixed input parameters from historical data sample themselves to get *in-the-sample* optimized performance – the *best possible* fixed asset allocation outcome during the historical sampling period.

For discrete *REDD-COPS*, current and historical portfolio values are used in the *forward* asset allocation weight calculation, even though the asset class parameters of λ , σ and ρ can still be estimated as constants from *in-the-sample* process. We call this historical data *Calibrated REDD-COPS* model. The purpose of *model calibration* is to demonstrate the value added by the active *forward* drawdown control mechanism, beyond what traditional asset allocation can do theoretically from *in-the-sample* parameter specification.

3.1 Single Risky Asset – May not be enough

For a period of 60.5 years ending 2011/6/30, we use *S&P 500 Total Return Index (SPTR)* as the single risky asset and 3-month *US T-bill* as the risk free asset to test a *Calibrated REDD-COPS*. At market close of the last trading day every month, the portfolio is rebalanced between *SPTR* and *T-bill* according to equation (5). The allocations are held through the end of next month then repeat the process. The σ and λ parameters in equation (5) are calculated from *SPTR* Index's monthly return streams for the same 60.5-year period (see Table 1 for values of σ and λ in the column for *SPTR*).

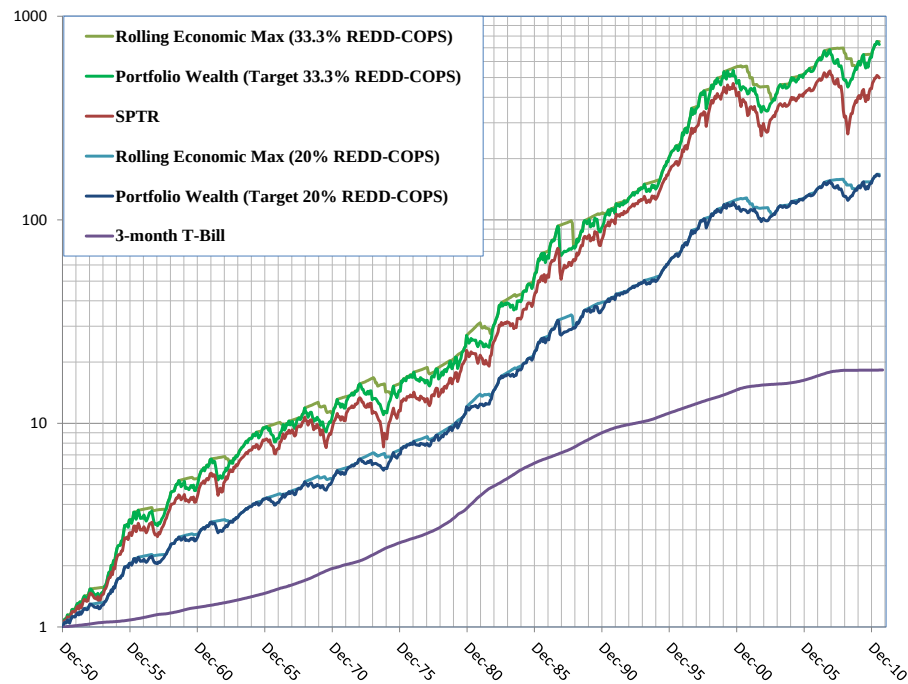
Table 1: *Calibrated REDD-COPS* of Single Risky Asset (*SPTR*): 1/1951-6/2011*

	SPTR	1/3 REDD-COPS	25% REDD-COPS	20% REDD-COPS	15% REDD-COPS
Annualized Return	10.81%	11.50%	9.79%	8.80%	7.82%
Annualized Std Deviation	14.61%	14.16%	10.03%	7.80%	5.73%
Max REDD	46.76%	29.43%	21.59%	17.10%	12.72%
Average REDD	6.19%	6.42%	4.58%	3.58%	2.63%
Max Drawdown	50.95%	37.67%	26.53%	19.89%	14.29%
Sharpe Ratio	0.403	0.465	0.486	0.497	0.506
Average Total Exposure	100%	103.74%	73.56%	57.32%	42.08%
Max Total Exposure	100%	122.30%	86.97%	67.94%	50.04%
Ending Multiple (60.5 years)	498.0	726.6	284.6	164.2	95.3
Skew	-0.418	-0.511	-0.504	-0.452	-0.444
Excessive Kurtosis	1.769	2.348	2.354	2.364	2.300

* In the first year 1951, *REDD* is evaluated by anchoring look-back to the end of 1950. The fixed rolling one-year look-back starts at the beginning of 1952. “Ending Multiple” refers to the portfolio dollar value on 2011/6/30 if it started as one dollar on 1950/12/31 without any deposit or withdrawal / tax during the 60.5 years. “Total Exposure” refers the portfolio portion x invested in risky asset *SPTR*. The “Average Total Exposure” and “Average *REDD*” reported are the simple arithmetic averages of the 726 monthly end values. Risk free rate for Sharpe ratio calculation is 4.92%, the average 3-month T-bill yield over the 60.5 years.

From Table 1, all “Max *REDD*” have been controlled within the target levels ($\delta = 1/3, 1/4, 1/5$, and 15%) tested, despite that the *SPTR* index deviates from the normal return distribution with negative skew and excess kurtosis. The *ex post* Maximum Drawdown stays within target δ for lower levels but is exceeded when $\delta \geq 1/4$. Compared to *SPTR*, *Calibrated REDD-COPS* with single risky asset ($\delta = 1/3$ in Table 1) reduces Max *REDD* and Max Drawdown substantially while adding returns.

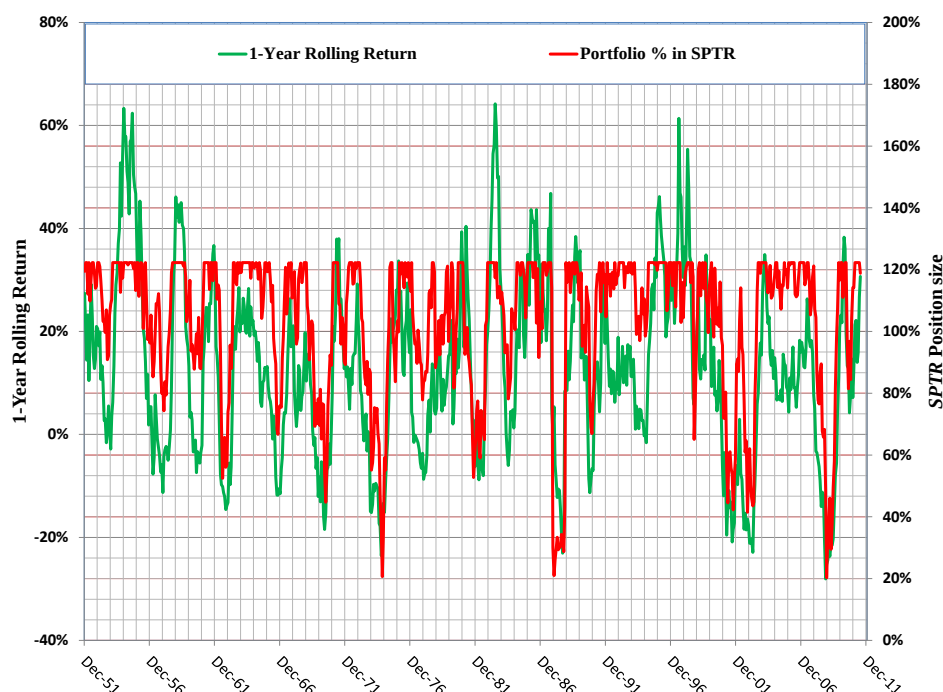
Figure 1: Growth of *Calibrated REDD-COPS* (Single Risky Asset *SPTR* 1951-2011/6)



But the level of *REDD* tolerance required ($\delta = 1/3$) is still high. Both the average and maximum exposure for the $\delta = 1/3$ case exceed 100%, indicating a leveraged (up to 22.3%) portfolio that must borrow in 3-month T-bill, or long S&P 500 Index futures, or use a leveraged index ETF. From the curve in Figure 1 where *Rolling Economic Max* is different from *portfolio wealth* level, we observe that the longest non-zero *REDD* lasted 3.5 years until June 2003, and maximum *REDD* of 29.4% occurred in Oct. 2008. Figure 2 shows the volatile cycle of risky asset allocation, and its high correlation (coefficient of 0.72) to the 12-month trailing returns. 13 of the 16 months with annual trailing losses more than 20% would have occurred during the 2002-2003 and 2008-2009 bear markets. We will mostly focus on the recent 20.5-year period (1990/12/31-2011/6/30) hereafter.

Figure 2: 12-month Rolling Returns and Monthly Allocation in SPTR

For Calibrated $\delta = 1/3$ REDD-COPS (12/31/1950 – 6/30/2011)



3.2 Effects of REDD-COPS Trading Frequency – Monthly rebalance sufficient

More frequent portfolio rebalancing for REDD-COPS increases transaction costs and may not improve performance, especially when the stock market index shows less one-step trending due to noises at a shorter time scale. We compare frequencies of monthly, weekly and daily rebalancing between SPTR Index and 3-month *T-bill*.

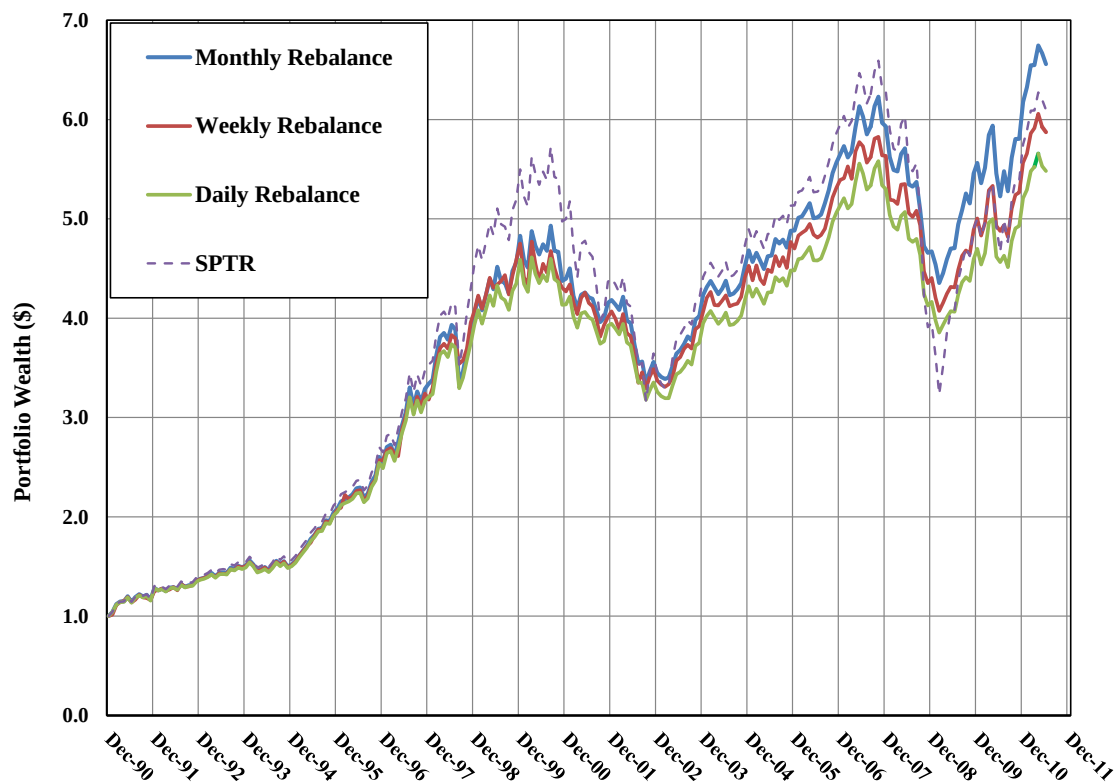
Table 2: Rebalance Frequency Effects*: $\delta = 30\%$ Calibrated REDD-COPS (SPTR/T-Bill)

	Annualized Return	Annualized Std Deviation	Sharpe Ratio	Maximum Drawdown	Maximum REDD	Average REDD	Average Exposure	Maximum Exposure
Monthly	9.61%	11.71%	0.521	32.11%	25.60%	4.90%	76.72%	101.53%
Weekly	9.02%	11.30%	0.488	31.01%	24.72%	4.86%	76.34%	101.53%
Daily	8.65%	11.09%	0.465	31.13%	25.41%	5.01%	75.16%	101.53%
SPTR	9.22%	14.89%	0.384	50.95%	46.76%	6.18%	100%	100%

* 20.5 year test period (from 1/2/1991 to 6/30/2011) is used. All portfolio risk metrics: standard deviation, simple drawdowns and REDD, are measured in discrete time intervals with *monthly* returns and month-end portfolio wealth levels, even when the rebalance trading is done daily or weekly. Regardless of rebalance frequencies, the input parameters, λ and σ , in equation (5) for SPTR are evaluated the same in the sample period from its monthly metrics listed in Table 3. One may choose to use the same data frequency of rebalance to calculate input λ and σ parameters. However daily or weekly measured volatility σ is more than that from monthly measurement, which leads to less leverage according to equation (5), but worse REDD-COPS performance.

The results are shown in Table 2 and Figure 3 for a 20.5 year (1991/01/2-2011/06/30) test period. Monthly rebalance outcome is shown to out-perform daily or weekly trading scheme in both returns and risk-adjusted returns. Active allocation of equation (5) has a 1-step momentum trading nature. Rising *SPTR* Index reduces *REDD* and raises *SPTR* allocation over next period, which will further gain if the uptrend continue. On the other hand, decline in *SPTR* leads to higher *REDD*, lower portfolio exposure or less losses should the downtrend continue. For the period 1/1991-6/2011, the monthly return time series of *SPTR* has a one-step-lag auto-correlation coefficient of 0.118, compared to weekly series' -0.081 and daily series' -0.062. Thus the out-performance of monthly rebalance can be explained by a stronger tendency of gain or loss to continue for the next period, compared to a weekly or daily rebalanced scheme.

Figure 3: Dollar Growth of $\delta = 30\%$ Calibrated *REDD-COPS* (*SPTR*/T-bill)



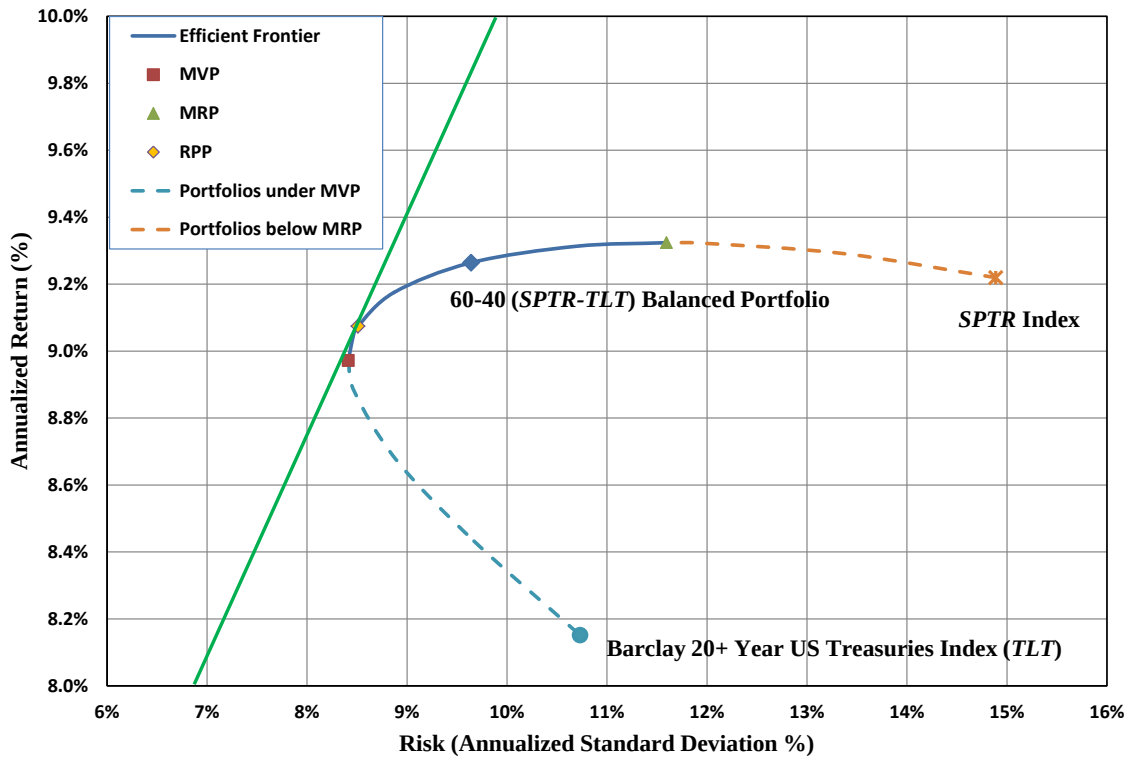
3.3 Two Risky Assets Case – Dominating Efficient Frontier

Over a 20.5-year test period, we use monthly data of *S&P 500 Total Return Index* (*SPTR*) and *Barclays Capital 20+ Year US Treasuries Index* (*TLT*) to calibrate *REDD-COPS* of two risky assets. The 3-month *T-Bill* is still used as the risk free asset.

Table 3: MPT Parameters for SPTR and TLT (Monthly data from 1/1991-6/2011)

	SPTR	TLT	Return Correlation (monthly)	Annualized risk free rate (average 3-month T-bills)
Annualized Return	9.22%	8.15%	$\rho = -0.069$	$r_f = 3.5\%$
Annualized Std Deviation	14.89%	10.73%		

Figure 4: Return vs. Risk: Fixed Allocation Portfolios of SPTR and TLT (1/1991-6/2011)



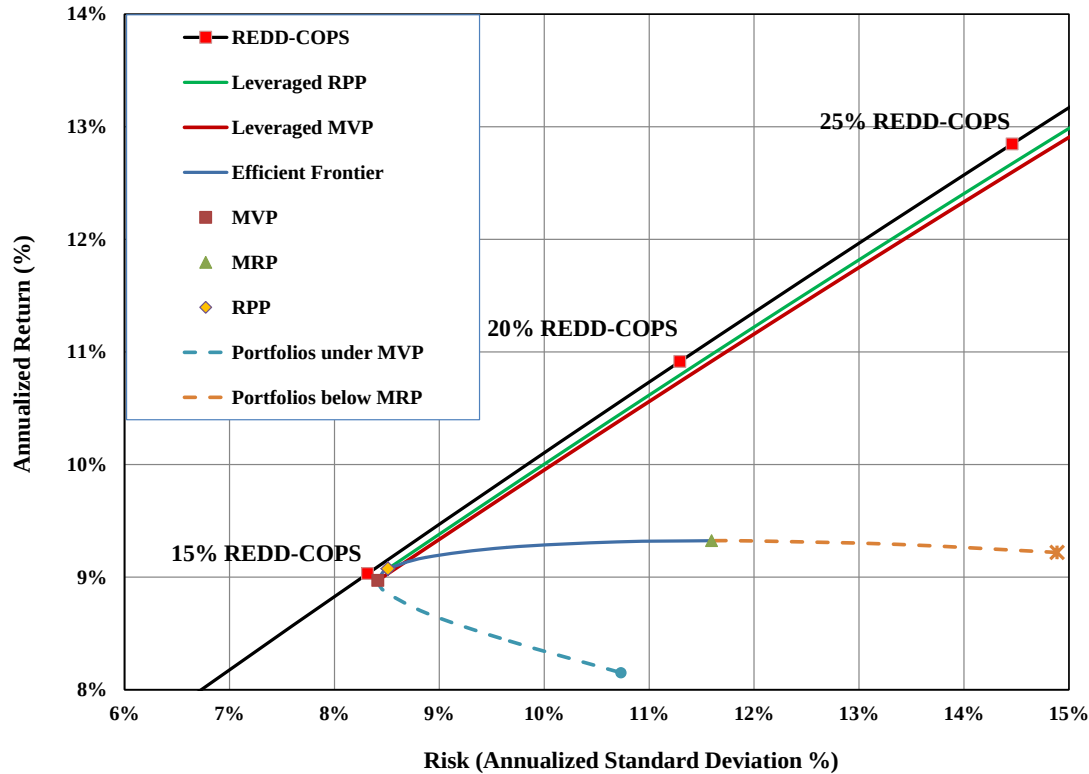
Efficient frontier of *SPTR* and *TLT* from traditional fixed asset allocations is plotted in Figure 4. With multi-period rebalance at end of every month, there is a **Maximum Return Portfolio (MRP)** of *SPTR* (77.3%) and *TLT* (22.7%) blend that achieved the highest annualized return among all fixed allocation portfolios over the 20.5 years. The portfolios between **Minimum Variance Portfolio (MVP)**⁵ and MRP form the efficient frontier (solid blue curve in Figure 4).

⁵ Two-asset fixed allocation MVP has portfolio weighting $w_{1,2} = \frac{\sigma_2^2 - \rho\sigma_1\sigma_2}{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2}$. When correlation

coefficient $\rho \leq 0$, $w_1 > 0$ and $w_2 > 0$ (no shorting) is guaranteed. For $1 > \rho > 0$, $\rho < \sigma_1/\sigma_2 < 1/\rho$ leads to no shorting. With *SPTR* and *TLT* index data over the recent 20.5 years (Table 3), MVP is a blend about 35.2% in *SPTR* and 64.8% in *TLT* that has a minimal level of 8.41% in annualized standard deviation (Table 4).

The two-asset **Risk Parity Portfolio (RPP)**⁶ on the efficient frontier turns out to be very close to the *tangential* portfolio for the example period. Thus, a blend of 3-month *T-bill* and *RPP* (green line in Figure 4) has better Sharpe ratio than any efficient frontier portfolio. The blend can be a *Leveraged Risk Parity Portfolio (LRPP)*^[10] that borrows 3-month *T-bill* for over 100% exposure in *RPP*.

Figure 5: Return-Risk Profile of Calibrated REDD-COPS (1/1991-6/2011)



Calibrated REDD-COPS are tested for wide range of drawdown target δ for the 20.5 years ending 6/30/2011. Figure 5 shows risk-return frontier of REDD-COPS (black line) with equation (3) used to define *REM* in REDD⁷. Leverage can also be applied for

⁶ The two-asset *RPP* has portfolio weighting as $w_{1,2} = \frac{\sigma_{2,1}}{\sigma_1 + \sigma_2}$. With parameters in Table 3 for *SPTR* and

TLT indexes for the recent 20.5 years, *RPP* is about 41.9% *SPTR* and 58.1% *TLT* blend, delivering 9.07% in annualized return and 8.51% of annualized standard deviation – a Sharpe ratio of 0.655. That is a better risk-adjusted return than the 0.598 Sharpe ratio of the 60-40 Balanced Portfolio (60% *SPTR* and 40% *TLT* blend - rebalanced monthly) for the same period. Table 4 lists performance statistics of *RPP*.

⁷ If the simpler equation (2) is applied to define *REM* value in equation (4), the REDD-COPS frontier with *SPTR/TLT* in the example period happens to be very close to match the *LRPP* frontier (Fig. 5's green line).

MVP and *RPP* by borrowing at risk free rate. Practically, leverage can be implemented through using T-bills as margin collateral and investing in index futures contracts of risky assets for more than 100% the nominal portfolio value. **Leveraged Risk Parity Portfolios** (*LRPP* – green line in Figure 5) and **Leveraged Minimum Variance Portfolios** (*LMVP* – brown line in Figure 5) are extensions from their baseline fixed allocation portfolios. *Calibrated REDD-COPS* frontier (black line in Figure 5) dominates *LRPP*, *LMVP* and Efficient Frontier from fixed allocation (blue line) in the risk-return space over 20.5 years. We summarize performance of the calibrated portfolios in Table 4 along with those for the *SPTR* and *TLT* indexes.

Table 4: Performance Statistics of Two Risky Assets (*SPTR* and *TLT*) based *Calibrated REDD-COPS* and Fixed Allocation Portfolios (1/1991-6/2011)

	Single Index Buy & Hold		<i>SPTR-TLT</i> Fixed Allocation Portfolio				<i>SPTR-TLT</i> REDD-COPS			
	<i>SPTR</i>	<i>TLT</i>	60/40 Portfolio (no-rebalance)	60/40 Portfolio (monthly-rebalance)	MVP	RPP	60% Levered RPP	15% REDD Target	20% REDD Target	25% REDD Target
Annualized Return	9.22%	8.15%	8.82%	9.26%	8.97%	9.07%	12.16%	9.03%	10.91%	12.85%
Annualized Std Deviation	14.89%	10.73%	10.47%	9.64%	8.41%	8.51%	13.58%	8.32%	11.29%	14.46%
Max REDD	46.76%	21.51%	27.14%	26.68%	15.67%	16.25%	26.04%	13.06%	17.65%	22.42%
Average REDD	6.18%	4.79%	4.11%	3.52%	2.92%	2.87%	4.63%	3.38%	4.59%	5.88%
Max Drawdown	50.95%	21.40%	30.30%	28.63%	15.65%	17.22%	28.16%	14.89%	20.45%	26.07%
Sharpe Ratio	0.384	0.433	0.508	0.598	0.650	0.655	0.638	0.665	0.656	0.646
Average Total Exposure	100%	100%	100%	100%	100%	100%	160%	93.99%	127.84%	163.80%
Max Total Exposure	100%	100%	100%	100%	100%	100%	160%	125.47%	170.35%	218.04%
Ending Multiple (20.5 years)	6.098	4.985	5.653	6.149	5.820	5.934	10.515	5.886	8.362	11.915

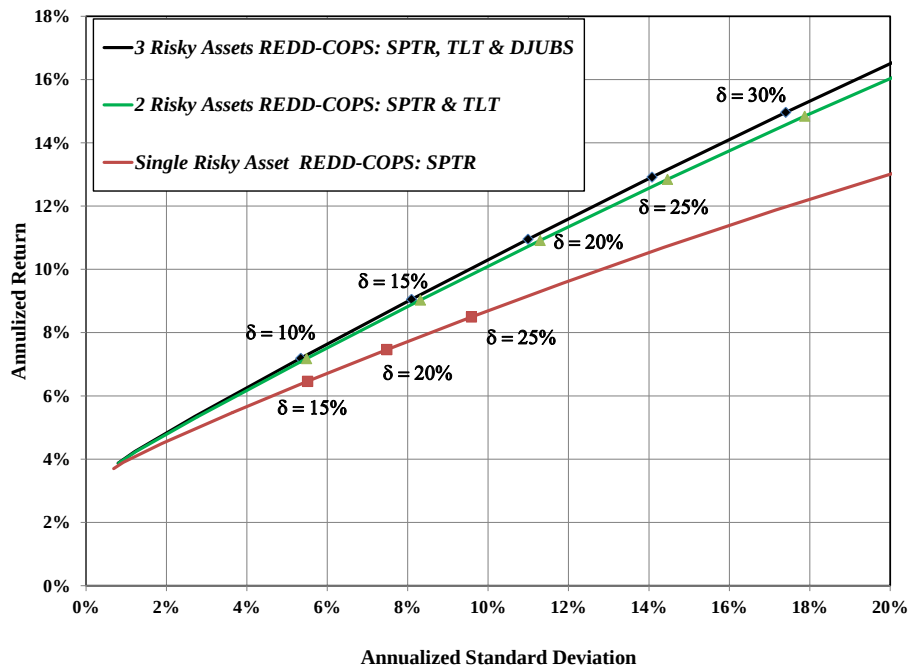
From equation (6), active trading for *Calibrated REDD-COPS* is driven by portfolio scaling factor of $\frac{\delta - REDD}{1 - REDD}$, superimposed to allocations in risky asset class indexes *SPTR* and *TLT*. Relative asset allocations in *SPTR* and *TLT* in *Calibrated REDD-COPS* are decided by input parameters λ and σ , the same as *MVP* and *LRPP* portfolios. The ratio between allocations to *SPTR* and *TLT* is $\frac{(\lambda_1 + \frac{1}{2}\rho\sigma_1 - \rho\lambda_2 - \frac{1}{2}\rho\sigma_2)/\sigma_1}{(\lambda_2 + \frac{1}{2}\rho\sigma_2 - \rho\lambda_1 - \frac{1}{2}\rho\sigma_1)/\sigma_2} = \frac{40.61\%}{59.39\%}$ from equation (6), very close to the asset allocation ratios between *SPTR* and *TLT* for *MVP* and *RPP* portfolios (see *Footnote 5* and *6*). Thus, the performance difference between *Calibrated REDD-COPS* and *MVP* or *LRPP* is mostly due to market timing of drawdown control to allocate between risky assets and risk free asset.

With similar portfolio's volatilities (annualized standard deviations), $\delta=15\%$ *REDD-COPS* returned more than *MVP* (comparing green columns in Table 4). $\delta=25\%$ *REDD-COPS* out-performed *60% leveraged RPP* with less Maximum Drawdown (comparing yellow columns in Table 4). Active trading through drawdown control mechanism in *Calibrated REDD-COPS* adds value in the example 20.5-year back test period.

3.4 Three Risky Assets Case – Additional Diversification Benefit

In addition to *SPTR* and *TLT* Indexes, we introduce into *Calibrated REDD-COPS* a third risky asset class index: *Dow-Jones UBS-Commodity Total Return Index (DJUBS)*. Despite the index's lower Sharpe ratio ($\lambda_{DJUBS} = 0.157$) over the 20.5 year period (since inception in 1/1991) than that of *SPTR* or *TLT*, it can bring diversification benefit and raise risk adjusted return when blending with a *SPTR-TLT* fixed allocation portfolio under Mean-Variance Optimization (*MVO*)⁸.

Figure 6: Return/Risk of *Calibrated REDD-COPS*: More Risky Assets (1/1991-6/2011)



⁸ We examine that a relation among Sharpe ratios and correlation coefficient of the *DJUBS* Index and the Tangential Portfolio (essentially *Risk Parity Portfolio* or *RPP* point on the *SPTR-TLT* efficient frontier in Figure 4), $\lambda_{DJUBS} > \rho_{DJUBS-RPP} \cdot \lambda_{RPP}$. This indicates that *DJUBS* is value adding under traditional *MVO* framework over the 20.5 year test period.

Over 20.5years, monthly return correlation coefficients among *DJUBS*, *SPTR* and *TLT* are low: $\rho_{DJUBS-SPTR} = 0.275$, $\rho_{DJUBS-TLT} = -0.085$ and $\rho_{SPTR-TLT} = -0.069$. Applying these correlation coefficients and the λ and σ parameters of *SPTR* and *TLT* (Table 4) and *DJUBS* (Table 5) in equation A.3 (Appendix 2), we can rebalance the *REDD-COPS* monthly among three risky asset class indexes *SPTR-TLT-DJUBS* and the risk free asset 3-month *T-bill*. Same as for the two risky assets case, *Calibrated REDD-COPS* with three risky-assets only applies active trading to scale overall position size in risky assets by a factor $\frac{\delta - REDD}{1 - REDD}$. The relative weightings in three risky asset indexes, however, are fixed. They have the ratios of: *SPTR* /*TLT* /*DJUBS* = $C_1/C_2/C_3 = 33.78\%: 54.86\%: 11.36\%$, where C_1 , C_2 and C_3 are calculated according to equation A.3 (Appendix 2).

Including *DJUBS* in *Calibrated REDD-COPS* portfolio improves portfolio efficiency to a better return-risk trade-off. As shown in Figure 6, under the same *REDD* control limit, adding *TLT* to the *SPTR* based *Calibrated REDD-COPS*, achieves significantly higher returns but also somewhat higher standard deviations. Adding *DJUBS* to the *SPTR-TLT* based *Calibrated REDD-COPS*, results in higher annualized returns and less portfolio return standard deviations when comparing Table 5 and Table 4. Results of *REDD-COPS* in Table 5 use equation (3) for *REM* calculation. Realized Maximum *REDD* breaks the drawdown control target slightly during 2009 in the $\delta = 25\%$ and 30% example cases.

Table 5: Performance Statistics of *Calibrated REDD-COPS* of Three Risky Assets and Buy-and-Hold *SPTR*, *TLT* or *DJUBS* Index (01/1991-06/2011)

	Single Index Buy & Hold			<i>SPTR-TLT-DJUBS</i> REDD-COPS				
	<i>SPTR</i>	<i>TLT</i>	<i>DJUBS</i>	10% REDD Target	15% REDD Target	20% REDD Target	25% REDD Target	30% REDD Target
Annualized Return	9.22%	8.15%	5.80%	7.19%	9.05%	10.95%	12.92%	14.96%
Annualized Std Deviation	14.89%	10.73%	14.70%	5.34%	8.10%	10.99%	14.08%	17.40%
Max REDD	46.76%	21.51%	54.48%	9.92%	14.93%	19.98%	25.09%	30.24%
Average REDD	6.18%	4.79%	7.34%	2.04%	3.09%	4.20%	5.37%	6.62%
Max Drawdown	50.95%	21.40%	54.26%	9.36%	14.46%	19.60%	24.77%	30.04%
Sharpe Ratio	0.384	0.433	0.157	0.690	0.686	0.678	0.669	0.659
Average Total Exposure	100%	100%	100%	71.07%	108.41%	147.67%	189.50%	234.69%
Max Total Exposure	100%	100%	100%	87.89%	133.53%	181.28%	232.04%	286.86%
Ending Multiple (20.5 years)	6.098	4.985	3.179	4.150	5.908	8.422	12.065	17.438

4. Risk-based Dynamic Asset Allocation to Control Drawdown

The *Calibrated REDD-COPS* using historical data has assumed three types of input parameters of the risky assets as constants: Sharpe ratios (λ), correlation coefficients (ρ) and return standard deviations (σ). We now revisit the input parameters issue to design *REDD-COPS* for practical implementation. Notice when correlation coefficients $\rho_{ij} = 0$, equations (6), A.1 and A.3 for multiple un-correlated risky assets are simplified. The allocation weights all reduce to the form of equation (5) for each risky asset. We can rewrite equation (5) for each un-correlated risky asset as:

$$x_i = \left(\frac{\bar{\lambda}_i}{\bar{\sigma}_i(t, h)} + \frac{1}{2} \right) \cdot \text{Max} \left(0, \frac{1}{1 - \delta^2} \cdot \frac{\delta - \text{REDD}(t, H)}{1 - \text{REDD}(t, H)} \right) \quad (7)$$

The allocation in risk free asset is still $x_f = 1 - \sum_{i=1}^n x_i$. Equation (7) is an easily implementable one-step forward trading rule: rebalance according to the portfolio value and *REDD* at time t , and hold for the time period $[t, t + \Delta t]$ before the next rebalance trade at time $(t + \Delta t)$. The assumptions about capital markets are:

1. We can find n investable asset class indexes that are un-correlated over long term;
2. Long term expected *Sharpe ratio* of each risky asset class index is stable and can be estimated as a constant in the dynamic asset allocation process of equation (7);
3. Each risky asset return follows random walk, but the return volatility σ_i is not constant and can be forecasted from returns during a preceding time window h .

In Appendix 3, these simplifications are examined with monthly market data in the last 36 years (ending 12/30/2011) on three asset class indexes: *SPTR*, *TLT* and Goldman Sachs Commodities Index (*GSCI* Total Return Index)⁹. Equation (7) is essentially a risk-based dynamic asset allocation scheme, as $\bar{\sigma}_i(t, h)$ is the volatility risk measure updated at

⁹ *DJUBS* index has only 21 years of historical data (1991-2011). It has high correlation at about 0.898 to the *GSCI* which has monthly data available since 1971.

time t from the most recent return series looking back time window h . Still, the dynamic asset allocation of $\left(\frac{\bar{\lambda}_i}{\sigma_i(t,h)} + \frac{1}{2}\right)$ is further scaled by factor $\frac{\delta - REDD(t,H)}{1 - REDD(t,H)}$, the result of active portfolio position sizing to control drawdown. Additionally, investor drawdown risk tolerance also weighs in as a leverage factor of $\frac{1}{1 - \delta^2}$. Similar to choosing look-back period H for $REDD$, we will use $h = 1$ year, i.e. 12-month trailing standard deviation to define the varying volatility risk in the examples.

We use $SPTR$, TLT and $DJUBS$ indexes to back-test for the recent 20 years (1/1992-12/2011). Based on historical statistics in Appendix 3, we use constant expected Sharpe ratios $\bar{\lambda}_{SPTR} = 0.4$, $\bar{\lambda}_{TLT} = 0.45$ and $\bar{\lambda}_{DJUBS} = 0.15$, and $\delta = 20\%$ as a baseline case. Close to actual Sharpe ratios in Table 5, relative high expected capital market growth is assumed. We call it *Risk-based High 20% REDD-COPS*.

Figure 7: Risk based High 20% REDD-COPS ($\bar{\lambda}_{SPTR} = 0.4$, $\bar{\lambda}_{TLT} = 0.45$, $\bar{\lambda}_{GSCI} = 0.15$)
Monthly Rolling 1-year Return & REDD (top) and Total Risk Exposure (bottom)

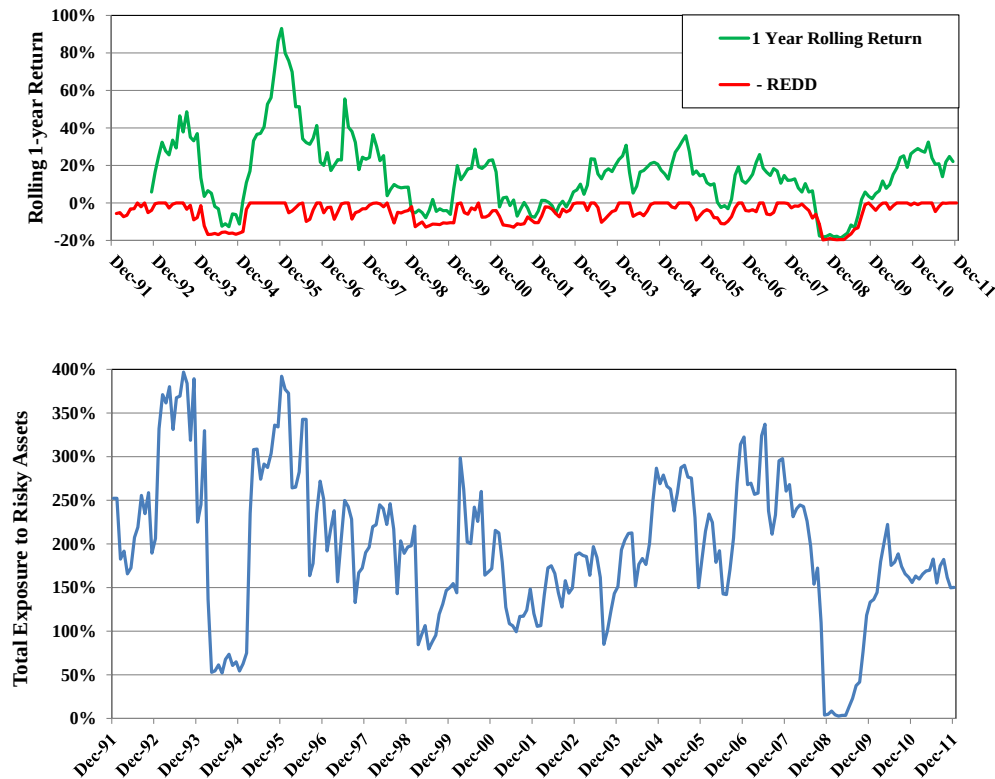
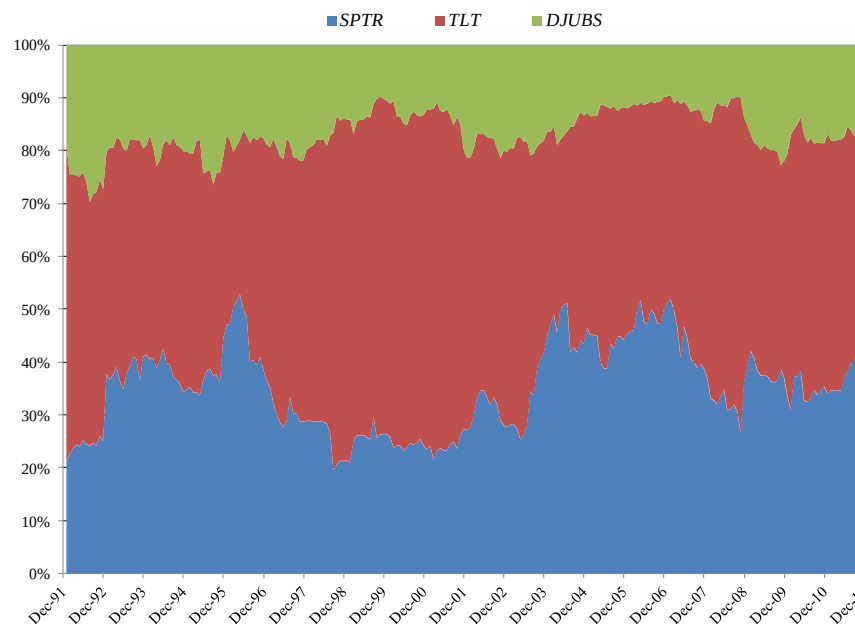


Figure 7 shows one year rolling return, *REDD*, and the portfolio total risk exposure in tandem. One-step momentum trading is evident: increase (decrease) in total exposure to risky assets follows the 1-year gain (loss) by one month. During 7 months (10/2008-4/2009) of the last bear market, *REDD-COPS* would have less than 5% total risk exposure. Overall, in 210 out of 240 months, the total portfolio exposure exceeded 100%. Maximum leverage was close to 300% in 07/1993. Monthly average leverage is about 100% over last 20 years.

Figure 8: Risk Based High 20% *REDD-COPS* (1992-2011): Distribution of Relative Portfolio Weights in Risky Assets (normalized to total 100%)



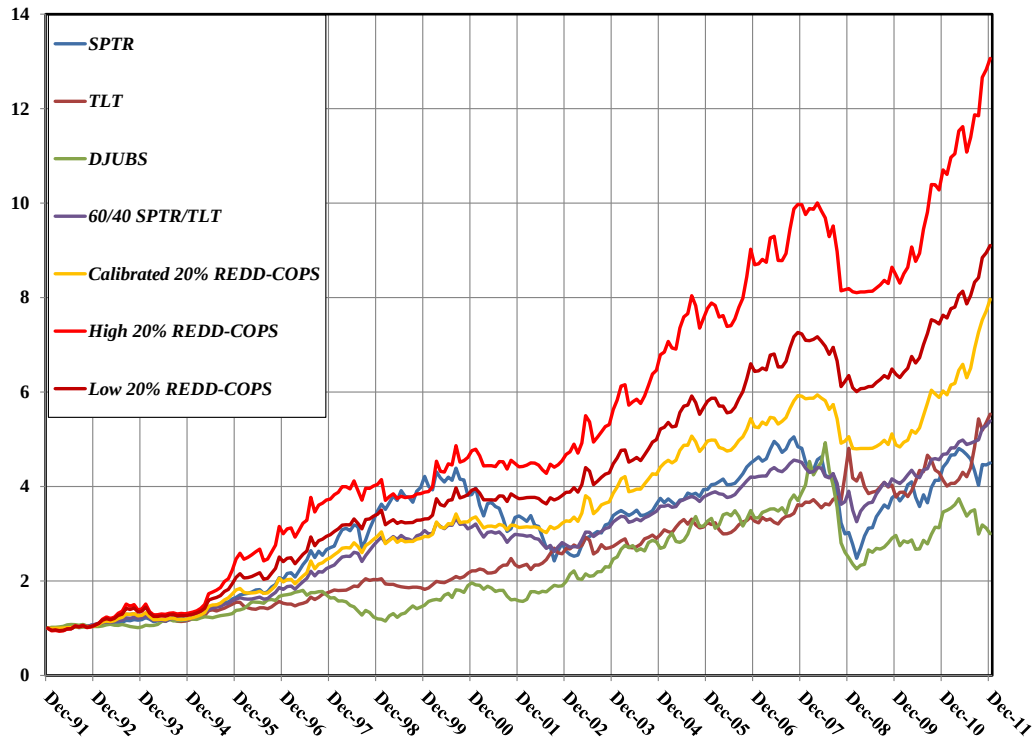
Within total risky asset exposure, distributions among the three indexes are shown in Figure 8 under a normalized sum to 100%. Over 20 years, temporal variations for relative allocations in each index are wide: *SPTR* (19.6% ~ 52.8%), *TLT* (29.2% ~ 66.5%) and *DJUBS* (9.5% ~ 29.7%). *REDD* drawdowns were successfully controlled within 20% over 20 years. *REDD* drawdown over 5% lasted 15 months in 7/2008 – 09/2009.

To address fluctuation of Sharpe ratios (see Appendix 3), we perturb each Sharpe ratio input in equation (6) by up to ± 0.15 from the baseline case, to examine *REDD-COPS* performance. For example, for capital market expectations of lower growth

referencing the 20-year Sharpe ratios of 1992–2011 (see Appendix 3), we use $\bar{\lambda}_{SPTR} = 0.3$, $\bar{\lambda}_{TLT} = 0.35$ and $\bar{\lambda}_{DJUBS} = 0.05$. At $\delta = 20\%$, we call it a *Low 20% REDD-COPS*. The only other parameter of equation (6) is the *REDD* target δ . We vary $\delta = 15\%$, 20% and 25% .

Figure 9 shows the risk-based *REDD-COPS* has the best growth in the last two decades while annual returns year by year are listed in Table 6. The *Calibrated REDD-COPS* uses asset allocation equation A.3 with constant parameters of expected index Sharpe ratios and volatilities (listed in Table 5), and correlation coefficients of Section 3.4. During the back-test of 20 years (1992–2011), the *risk-based High 20% REDD-COPS* out-performed the *Low 20% REDD-COPS* and the *Calibrated REDD-COPS*.

Figure 9: Dollar Growth of Risk Based 20% *REDD-COPS* and Benchmarks (1992–2011)



From return risk trade-off, risk-based *REDD-COPS* dominate *Calibrated REDD-COPS* as shown in Figure 10. It also demonstrates the robustness of risk-based *REDD-COPS*: higher capital market expectation makes little difference to the portfolio efficiency between return and standard deviation or monthly average *REDD*, but does lead to higher return for the same level of *ex post* Maximum *REDD* or Drawdown.

Table 6: Annual Returns: Risk-based 20% *REDD-COPS* and Benchmarks (1992-2011)

	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001
<i>SPTR</i>	7.62%	10.08%	1.32%	37.58%	22.96%	33.36%	28.58%	21.04%	-9.10%	-11.89%
<i>TLT</i>	7.93%	18.00%	-8.22%	32.59%	-1.46%	16.18%	14.22%	-10.06%	21.50%	3.63%
<i>DJUBS</i>	3.70%	-1.07%	16.61%	15.21%	23.17%	-3.39%	-27.03%	24.35%	31.84%	-19.51%
<i>60/40 SPTR/TLT</i>	7.81%	13.26%	-2.53%	35.68%	12.73%	26.45%	23.63%	7.76%	2.45%	-5.23%
Calibrated <i>REDD-COPS</i>	8.27%	21.32%	-5.69%	50.66%	9.41%	25.86%	18.23%	-0.06%	12.73%	-5.70%
High <i>REDD-COPS</i>	5.85%	33.11%	-6.23%	86.45%	21.80%	24.42%	8.37%	-4.06%	22.62%	-7.20%
Low <i>REDD-COPS</i>	6.59%	27.82%	-6.09%	62.10%	16.19%	23.66%	14.55%	-2.91%	18.35%	-4.58%
	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011
<i>SPTR</i>	-22.10%	28.68%	10.88%	4.91%	15.79%	5.49%	-37.00%	26.46%	15.06%	2.11%
<i>TLT</i>	17.00%	1.80%	8.99%	8.57%	0.93%	10.15%	33.72%	-21.40%	9.38%	33.84%
<i>DJUBS</i>	25.91%	23.93%	9.15%	21.36%	2.07%	16.23%	-35.65%	18.91%	16.83%	-13.32%
<i>60/40 SPTR/TLT</i>	-7.26%	17.70%	10.28%	6.60%	9.74%	7.63%	-13.59%	5.34%	14.01%	15.06%
Calibrated <i>REDD-COPS</i>	4.56%	17.80%	14.25%	12.62%	6.06%	12.31%	-14.49%	-3.47%	23.45%	32.20%
High <i>REDD-COPS</i>	5.94%	20.27%	20.65%	14.46%	11.87%	14.64%	-17.81%	3.53%	26.20%	22.07%
Low <i>REDD-COPS</i>	3.57%	15.61%	16.42%	11.26%	10.89%	12.36%	-12.20%	0.46%	19.58%	19.33%

Figure 10: Risk-Return of Risk-based *REDD-COPS* (1992-2011) $\delta = 2\%$ to 30%

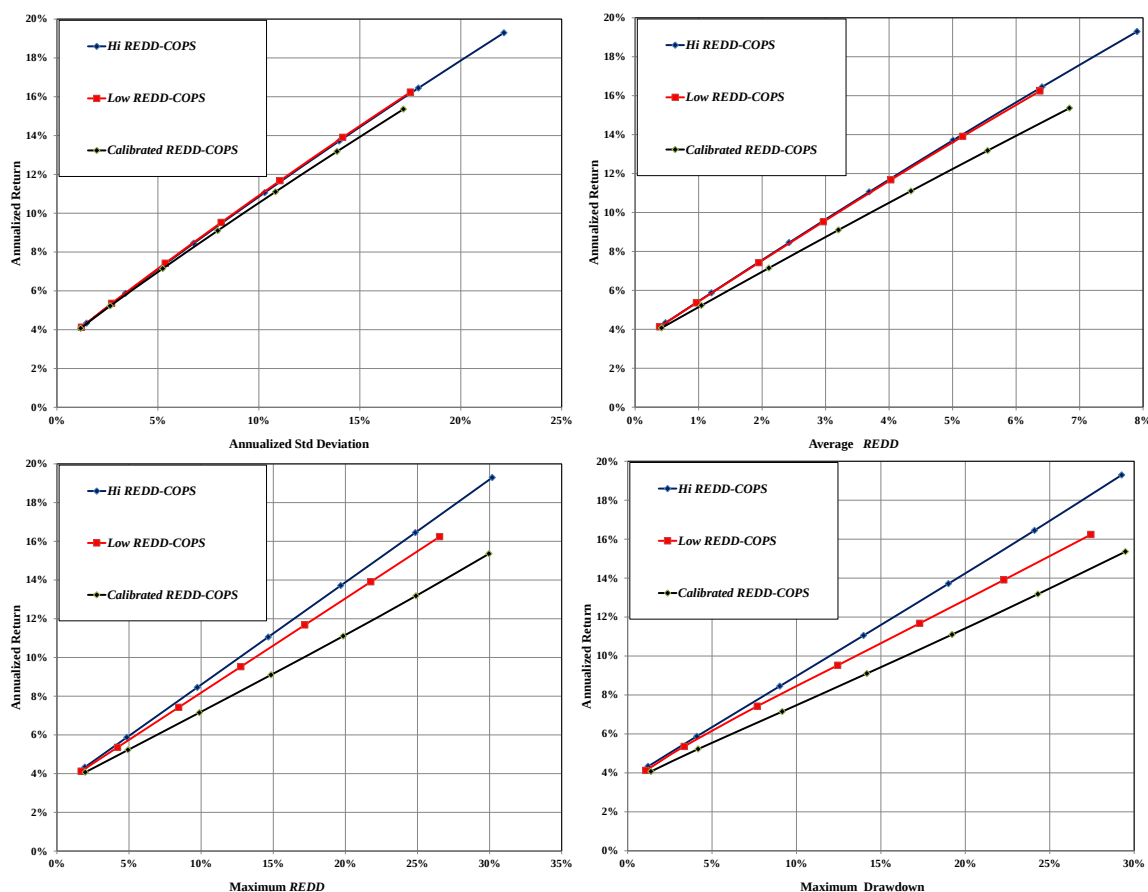


Table 7: Risk-based 20% *REDD-COPS*: Robustness to Market Expectations (1992-2011)

<i>Sum of 3 Expected λ's</i>	1	0.7	1.15	1.15	1.15	1.05	1.05	1	1	1	1	0.9	0.85	0.85
<i>Expected λ SPTR</i>	0.4	0.30	0.4	0.4	0.55	0.4	0.55	0.25	0.25	0.4	0.55	0.4	0.25	0.4
<i>Expected λ TLT</i>	0.45	0.35	0.6	0.45	0.45	0.6	0.45	0.45	0.6	0.3	0.3	0.45	0.45	0.3
<i>Expected λ DJUBS</i>	0.15	0.05	0.15	0.3	0.15	0.05	0.05	0.3	0.15	0.3	0.15	0.05	0.15	0.15
Annualized Return	13.71%	11.89%	14.70%	14.67%	14.60%	14.28%	14.07%	13.71%	13.65%	13.31%	13.19%	13.25%	12.65%	12.44%
Annualized Std Dev	13.98%	11.42%	15.63%	14.97%	15.54%	15.01%	15.03%	13.50%	14.42%	13.68%	14.18%	13.30%	12.49%	12.43%
Sharpe Ratio	0.745	0.752	0.730	0.759	0.727	0.731	0.717	0.771	0.718	0.732	0.698	0.748	0.749	0.735
Average REDD	5.01%	4.06%	5.82%	5.34%	5.59%	5.81%	5.54%	4.99%	5.49%	4.86%	5.24%	4.95%	4.51%	4.44%
Max REDD	19.68%	18.10%	20.05%	20.64%	20.76%	19.32%	20.32%	19.45%	18.62%	20.31%	20.43%	19.01%	18.13%	19.41%
Max Drawdown	19.00%	17.44%	19.29%	20.05%	20.52%	19.88%	21.25%	18.63%	17.89%	19.72%	21.50%	19.46%	17.40%	19.03%
Maximum Exposure	396.8%	305.6%	450.3%	464.2%	453.6%	425.5%	419.2%	402.6%	396.0%	422.7%	412.2%	367.0%	342.4%	350.6%
Average Exposure	195.2%	161.2%	211.8%	215.8%	210.7%	196.1%	195.1%	196.7%	193.7%	194.3%	188.6%	178.9%	176.6%	173.1%
Ending Multiple	13.065	9.458	15.537	15.444	15.275	14.430	13.918	13.055	12.927	12.183	11.919	12.034	10.831	10.437

The example *Risk-based REDD-COPS* appears robust against variations of capital market expectations, i.e., the input parameters of expected asset class Sharpe ratios. Table 7 indicates stable *ex post* risk-adjusted returns from risk-based *REDD-COPS*: Sharpe ratios ranges only from 0.698 to 0.759 over the 20-year period, despite that sum of three input index Sharpe ratios varies widely from 0.7 to 1.15. Larger total expected Sharpe ratios input does result in higher returns and higher “Average REDD” due to larger “Average Exposure” in risky assets. Some cases of the *Risk based 20% REDD-COPS* in Table 7 lose control of Maximum REDD within the 20% target despite the extent of “violation” is less than 1%. This is due to higher expected Sharpe ratios in riskier assets *SPTR* and *DJUBS*, or larger sum of three Sharpe ratios input. The *Low 20% REDD-COPS* (sum of input Sharpe ratios at 0.7) still has about 2% margin of safety to reach the *Max REDD* or *Max Drawdown* limit of 20% during the 2008-2009 bear market.

5. Discussion

The drastic market cycle of sharp decline during the 2008-2009 financial crisis and subsequent strong recovery in 2009-2011, provides us the opportunity to demonstrate that the continuous drawdown controlled optimal portfolio theory of Grossman and Zhou (1993) loses optimality under a straightforward discretized rebalancing extension. Our proposal of *Risk-based REDD-COPS* depends on a shorter portfolio history by looking back only a constant rolling time window. In what is essentially one-step momentum trading at discrete time intervals, *REDD-COPS* avoids a cascading performance drag due to the long memory of anchored drawdown, especially over a sharp V-shape market cycle.

Intuitively, the *REDD-COPS* can time the market well when the *market drawdown cycle* (from peak to bottom) is somewhat longer than (or at least matches) the rolling time window defining *REDD*. Right before a market rally from low, a lower *Economic Max* in the shorter time window results in larger allocation in risky assets to boost subsequent returns. In empirical tests with market index data, we have used *one year* as the rolling drawdown time window. During the 2008-2009 market crash, the *drawdown cycle* lasted 17 months for S&P 500 Index (*SPTR*) after its Oct. 2007 high, and 8 months for the Dow Jones UBS Commodities Index (*DJUBS*) after its June 2008 high, until the March 2009 low. One major factor preventing further decline was the coordinated market intervention from the world's central banks. Subsequent liquidity injections from the Federal Reserve also lifted US markets through mid-2011. Looking forward, a "Fed put" or strong external support after a major crash, can avoid an immediate "second leg down", and more probably lead to a strong rebound. This mitigates the risk of shorter memory in our opportunistic *REDD* control mechanism. Although the test examples use constant look-back window to define *REDD*, a variable time window can be optimal due to macro-economic cycle, monetary policy and speed of market impact from excess liquidities, etc.

Due to stronger serial correlation, monthly trading of *SPTR* in *REDD-COPS* has better performance than weekly or daily rebalancing. Monthly rebalance makes transaction cost less a concern¹⁰ and we use it extensively in *REDD-COPS* examples.

The three asset class indexes, *SPTR*, *TLT* and *DJUBS*, used in our back-test examples have nearly zero correlations to each other overall in the past 20+ years. In practice, one can perform *Principal Component Analysis (PCA)* over a pool of asset class indexes' historical return series to find leading principal factors that are uncorrelated. Risk-based active trading scheme of equation (7) can be applied more rigorously to the investable principal factors, each of which is a combination of underlying asset class indexes.

¹⁰ Since 2010, some North American brokerages (e.g. TD Ameritrade), have allowed monthly trading of broad index ETF's (like *IVV*, *TLT*, *DBC* and *BIL*) free of transaction fee for retail accounts. This makes the *REDD-COPS* with lower drawdown limit (<15%) practically possible for smaller account sizes.

Table 8: Performance of Risk-based *High REDD-COPS* versus Benchmarks (1992-2011)

	Single Index Buy & Hold			SPTR-TLT Fixed Allocation Portfolio			SPTR-TLT-DJUBS Risk-based REDD-COPS		
	SPTR	TLT	DJUBS	Balanced (60-40)	MVP	60% Levered RPP	15% REDD Target	20% REDD Target	25% REDD Target
Annualized Return	7.81%	8.93%	5.65%	8.79%	9.04%	11.29%	11.06%	13.71%	16.45%
Annualized Std Deviation	15.00%	11.37%	15.23%	9.46%	8.40%	13.35%	10.29%	13.98%	17.90%
Sharpe Ratio	0.301	0.495	0.154	0.580	0.683	0.598	0.754	0.745	0.734
Average REDD	7.21%	5.64%	8.01%	4.31%	3.68%	5.45%	3.69%	5.01%	6.41%
Max REDD	46.76%	21.51%	54.48%	26.68%	20.51%	26.04%	14.65%	19.68%	24.85%
Max Drawdown	50.95%	21.40%	54.26%	28.63%	19.09%	28.16%	13.98%	19.00%	24.10%
Maximum Exposure	100%	100%	100%	100%	100%	160%	292.3%	396.8%	507.9%
Average Exposure	100%	100%	100%	100%	100%	160%	143.3%	195.2%	250.3%
Ending Multiple	4.501	5.533	3.001	5.389	5.641	8.489	8.143	13.065	21.014

Theoretically, the optimal relative asset allocation weights in *REDD-COPS* formula ($[\Sigma^{-1} \cdot \mu]^T \cdot \Sigma^{-1}$ portion in equation A.1) is the same as the unconstrained investor wealth utility optimization extension from Modern Portfolio Theory (*MPT*). We introduce active management into the relative asset allocation weights by allowing the inverse of each asset class volatility to change at portfolio trading frequency looking back a rolling period (e.g. $h = 12$ months). The rest of the relative asset allocation variables among uncorrelated asset classes are simply their expected Sharpe ratios. We choose to specify them as input parameters using long term historical estimations.

Traditional asset allocation process based on *MPT* struggled with sensitivities to parameter inputs. For a wide range variation of constant Sharpe ratio parameter inputs, performance of the *Risk based REDD-COPS* examples (with *SPTR-TLT-DJUBS* indexes) is robust over 20 years (1992-2011). With the same *REDD* drawdown control mechanism, efficient frontier of the *Risk-based REDD-COPS* even dominates that of the *Calibrated REDD-COPS* which has constant volatility and correlation inputs from *in-the-sample* estimations. This indicates that the dynamic asset allocation using active volatility estimates adds value in the back-test period. Still, majority of the market timing gains can be attributed to the active position sizing factor $\frac{\delta - REDD}{1 - REDD}$ for *REDD* drawdown control.

Table 8 summarizes the performance comparison of *Risk-based REDD-COPS* ($\bar{\lambda}_{SPTR} = 0.4$, $\bar{\lambda}_{TLT} = 0.45$ and $\bar{\lambda}_{DJUBS} = 0.15$) against traditional asset allocation portfolios and indexes. In general, maximum drawdown or *REDD* are successfully controlled within target limit.

We find relative high leveraged level of exposure in risky asset class indexes are needed for *Risk-based REDD-COPS* implementation. Cost of borrowing for leveraged investment in risky asset class indexes is assumed at risk free rate. In reality, leverage can reduce the portfolio returns as funding cost can be higher than 3-month *T-Bill* rate. Time decay in derivative prices can degrade performance even if the leverage is realized through index futures contract or long dated index options. Still the *Risk-based REDD-COPS*' margin of out-performance is substantially high. For example, about 5% in annualized excess return of *Risk-based REDD-COPS* over a passive asset allocation benchmark portfolio with similar maximum drawdown loss, is achieved for the test period 1992-2011(see Table 8), much more than the potential cost of market friction from extra leverage transactions. In practice, a retail investor can implement the example *risk-based REDD-COPS* of lower drawdown target ($\delta \leq 15\%$) with leveraged index *ETF* and mutual funds. For institutional investors or managers of global macro or managed futures strategies, *REDD-COPS* with higher drawdown limit ($\delta \geq 20\%$) can be considered given the access to higher leverage through derivative markets.

6. Conclusion

At the highest level of investment management, asset allocation is responsible for most of the portfolio performance^[11]. As an active asset allocation framework, our *Risk-based REDD-COPS* proposed three new methodologies for practical portfolio management: 1. *Rolling Economic Drawdown (REDD)* for portfolio loss control; 2. Risk aversion defined as the dynamic floor of maximum loss tolerance; 3. Dynamic asset allocation implemented through the inverse of asset class volatility estimation.

Besides the benefit of a clear mandate to construct suitable client portfolios, the methodologies of *REDD-COPS* can be applied to design invest-able securities, such as a principal-guaranteed investment product, target risk *ETF*, and target-date asset allocation mutual fund with a glide path, etc. With active trading mechanisms derived solely from risk statistics and risk target, the current proposal of *REDD-COPS* demonstrates the prowess of risk based investment management: driven by active risk control, rather than performance chasing, the portfolio will out-perform, consistently, for the long term.

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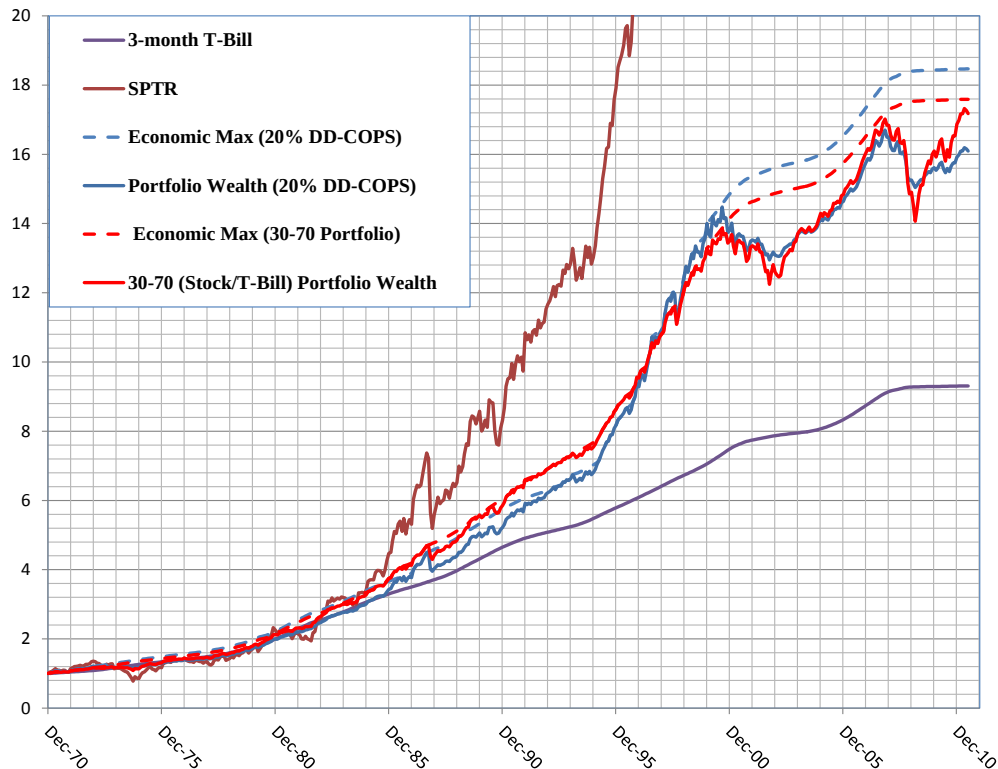
Appendix 1

Historical Data Demonstration: Discrete DD-COPS loses Optimality

Under the same constraint as discrete *DD-COPS* that *Economic Drawdown (EDD)* stays below a target level δ , another tradable allocation scheme is needed to show better long term accumulated return. Here is the set up:

- Use *S&P 500 Total Return Index (SPTR)* as the single risky asset, and 3-month US *T-bill* as the risk free bond. Take $\delta = 20\%$ as the target of *EDD* constraint.
- The reference alternative portfolio assumes a fixed allocation, 30% in *SPTR* and 70% in 3-month *T-bill*. Call it the *30-70 (Stock/T-Bill) Portfolio*.
- Both *DD-COPS* and *30-70 (Stock/T-Bill) Portfolio* uses monthly rebalance, trades at market close value of *SPTR* Index on last trading day of each month.
- The time span is from 1/2/1971 to 6/30/2011. In equation (1), the long term return R , standard deviation σ and risk free rate r_f , are constant parameters evaluated over the period (see Table A1).

Figure A.1: Growth of 20% *DD-COPS* vs. 30/70 *Stock/T-Bill* Portfolio for 40.5 Years



From the above Figure, *DD-COPS* did not make any new *Economic Max* after 2000. Even when *DD-COPS* value reached a new high in late 2007, it is still below the *T-bill* yield compounded high watermark from year 2000. Although the finite *EDD* extended into the 2008-2009 market decline helped to limit the *Max Drawdown* to just 12%, it further reduced stock index exposure during 2009-2011 when S&P 500 Index rose sharply. As a result of the cascading effect, the *DD-COPS* never made a new high in 2011 while the *30/70 Stock/T-bill portfolio* did.

Table A1: 20% *DD-COPS* vs. the *30-70 Stock/T-Bill Portfolio* and *SPTR* Index
Performance Statistics for 40.5 Years (1971-2011/6/30)

	<i>SPTR</i>	<i>30-70 Stock/T-Bill</i>	20% Target <i>DD-COPS</i>
Annualized Return	10.18%	7.27%	7.10%
Annualized Std Deviation	15.51%	4.69%	4.89%
Max <i>EDD</i>	55.65%	19.77%	18.31%
Average <i>EDD</i>	15.59%	4.39%	8.41%
Max Drawdown	50.95%	17.35%	12.01%
Sharpe Ratio	0.291	0.343	0.294
Average Total Exposure	100%	30%	30.61%
Max Total Exposure	100%	30%	49.50%
Ending Multiple (40.5 years)	50.69	17.18	16.10
Skew	-0.448	-0.454	-1.037
Excessive Kurtosis	1.996	2.135	7.620

As shown in Table A1, the 40-year annualized return and “ending multiples” from the 20% target *DD-COPS* turns out to be less than that from the referenced fixed allocation scheme of the *30-70 Stock/T-Bill Portfolio*. This example thus demonstrates a situation that *DD-COPS* loses optimality in the long term portfolio growth rate.

Appendix 2

Derivation of Multiple Risky Assets REDD-COPS Allocation Weights

Taking $\delta = \gamma$ in Cvitanic and Karatzas (1994)^[2], the column vector $[x]$ of risky asset allocation weights of an REDD-COPS with multiple risky assets, is given by:

$$[x]^T = \left([\Sigma^{-1} \cdot \mu]^T \cdot \Sigma^{-1} \right) \cdot \max \left(0, \frac{1}{1-\delta^2} \cdot \frac{\delta - REDD}{1 - REDD} \right) \quad (A.1)$$

where the matrix Σ is the lower triangular Cholesky decomposition of the return covariance matrix $\Omega = [\rho_{ij}\sigma_i\sigma_j] = \Sigma \cdot \Sigma^T$. The μ vector is the return drift of risky assets that each component $\mu_i = R_i - r_f + \frac{1}{2}\sigma_i^2$ where R_i is i 'th risky asset's long term return.

We have $\Sigma = \begin{bmatrix} \sigma_1 & 0 \\ \rho\sigma_2 & \sqrt{1-\rho^2}\sigma_2 \end{bmatrix}$, $\Sigma^{-1} = \begin{bmatrix} 1/\sigma_1 & 0 \\ -\rho/(\sigma_1\sqrt{1-\rho^2}) & 1/(\sigma_2\sqrt{1-\rho^2}) \end{bmatrix}$ for case of two risky assets. Substitute them into equation A.1 for $\begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$. With $\lambda_{1,2} = \frac{R_{1,2} - r_f}{\sigma_{1,2}}$, it yields eqn. (6):

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \frac{1}{1-\rho^2} \begin{bmatrix} (\lambda_1 + \frac{1}{2}\sigma_1 - \rho \cdot \lambda_2 - \frac{1}{2} \cdot \rho \cdot \sigma_2)/\sigma_1 \\ (\lambda_2 + \frac{1}{2}\sigma_2 - \rho \cdot \lambda_1 - \frac{1}{2} \cdot \rho \cdot \sigma_1)/\sigma_2 \end{bmatrix} \cdot \text{Max} \left[0, \frac{1}{1-\delta^2} \cdot \left(\frac{\delta - REDD}{1 - REDD} \right) \right]$$

For three-dimensional case (three risky assets), we have:

$$\Sigma = \begin{bmatrix} \sigma_1 & 0 & 0 \\ \sigma_2\rho_{12} & \sigma_2\sqrt{1-\rho_{12}^2} & 0 \\ \sigma_3\rho_{13} & \sigma_3(\rho_{23} - \rho_{12}\rho_{13})/\sqrt{1-\rho_{12}^2} & \sigma_3\sqrt{1-\rho_{13}^2 - (\rho_{23} - \rho_{12}\rho_{13})^2/(1-\rho_{12}^2)} \end{bmatrix} \quad (A.2)$$

We can derive Σ^{-1} and substitute into equation A.1, after some algebra, we have:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \frac{1}{1-\theta^2} \cdot \begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} \cdot \text{Max} \left[0, \frac{1}{1-\delta^2} \cdot \left(\frac{\delta - REDD}{1 - REDD} \right) \right] \quad (A.3)$$

$$\theta^2 = \rho_{12}^2 + \rho_{23}^2 + \rho_{31}^2 - 2\rho_{12}\rho_{23}\rho_{13}$$

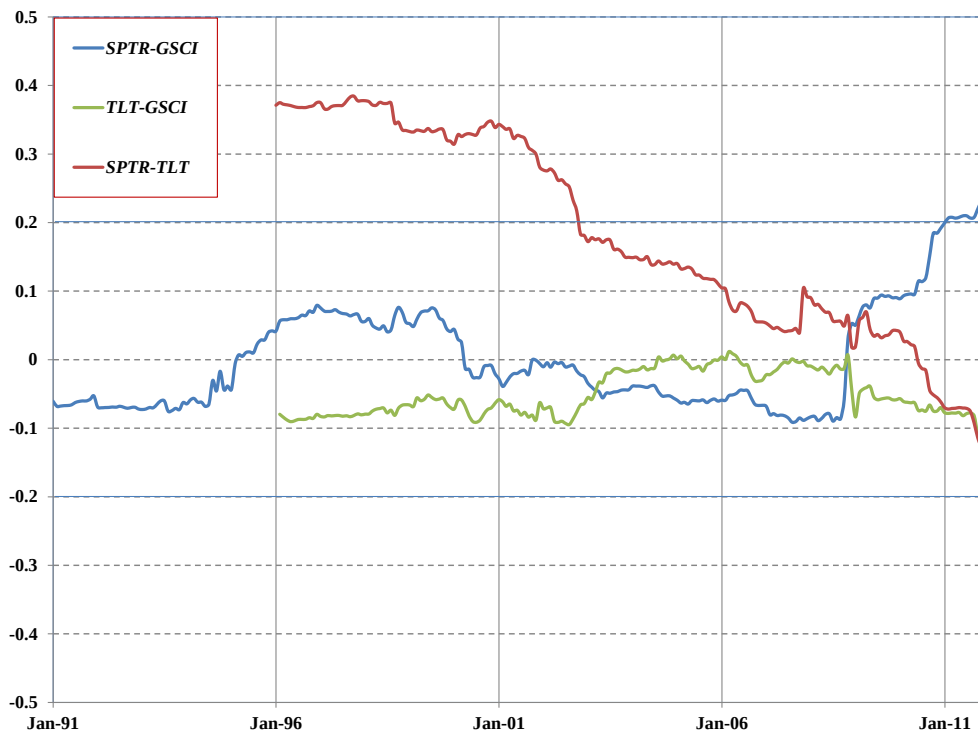
$$\begin{bmatrix} C_1 \\ C_2 \\ C_3 \end{bmatrix} = \begin{bmatrix} \left\{ (\lambda_1 + \frac{1}{2}\sigma_1)(1-\rho_{23}^2) + (\lambda_2 + \frac{1}{2}\sigma_2)(\rho_{23}\rho_{13} - \rho_{12}) + (\lambda_3 + \frac{1}{2}\sigma_3)(\rho_{23}\rho_{12} - \rho_{13}) \right\} / \sigma_1 \\ \left\{ (\lambda_2 + \frac{1}{2}\sigma_2)(1-\rho_{13}^2) + (\lambda_1 + \frac{1}{2}\sigma_1)(\rho_{13}\rho_{23} - \rho_{12}) + (\lambda_3 + \frac{1}{2}\sigma_3)(\rho_{13}\rho_{12} - \rho_{23}) \right\} / \sigma_2 \\ \left\{ (\lambda_3 + \frac{1}{2}\sigma_3)(1-\rho_{12}^2) + (\lambda_1 + \frac{1}{2}\sigma_1)(\rho_{23}\rho_{12} - \rho_{13}) + (\lambda_2 + \frac{1}{2}\sigma_2)(\rho_{13}\rho_{12} - \rho_{23}) \right\} / \sigma_3 \end{bmatrix}$$

Appendix 3

Historical Correlation and Sharpe Ratios on Three Asset Class Indexes

We take a rolling window of look-back 20 years and compute monthly return correlation coefficients at the end of each month. There are three pairs among three asset class indexes: *SPTR*, *TLT* and *GSCI* – all three using the total return index series. Since the *TLT* index monthly return data started the latest of the three in Jan 1976, we only have this 20-year rolling correlation for the past 16 years ending Dec. 2011. See Figure A.2 below.

Figure A.2: History of the Rolling 20-year Return Correlation Coefficients



During 16 years, the arithmetic averages of the 20-year trailing correlation coefficients are $\bar{\rho}_{SPTR-GSCI} = 0.019$, $\bar{\rho}_{SPTR-TLT} = 0.181$ and $\bar{\rho}_{TLT-GSCI} = -0.051$. The correlation coefficients are also bounded between -0.2 and 0.4 at all sampled time. So for a 20-year window, the un-correlated or low correlation assumption about *SPTR*, *TLT* and *GSCI* indexes is held reasonably well in historical observations.

We take the same approach of 20-year look-back to compute Sharpe ratios of the three indexes at the end of each month. The risk free rate from 3-month *T-bill* is also computed from the compounded return during the 20-year window. Risky asset class index's return (R) and standard deviation (σ), and risk free rate (r_f) are all annualized to compute Sharpe ratio (λ). See Figure A.3 below for the 21-year history of the Sharpe ratios of *SPTR* and *GSCI* indexes and 16- year history of *TLT* index Sharpe ratio. The averages are $\bar{\lambda}_{SPTR} = 0.446$ (on 252 monthly samples) $\bar{\lambda}_{TLT} = 0.432$ (on 192 monthly samples) and $\bar{\lambda}_{GSCI} = 0.166$ (on 252 monthly samples). Notice the fluctuations and range for Sharpe ratios. Thus we also compute the standard deviations of the Sharpe ratios from these samples. $\sigma_{\lambda-SPTR} = 0.146$, $\sigma_{\lambda-TLT} = 0.131$, $\sigma_{\lambda-GSCI} = 0.127$.

Figure A.2 Histories of 20-year Rolling Asset Class Index Sharpe Ratios

